An Empirical Evaluation of the GPT-4 Multimodal Language Model on Visualization Literacy Tasks

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Reference

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Citations

10

Number of Pages

11

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Why we chose this paper?

- **Personal interest**
- **Advances our understanding of AI in visualization**
- **Relevance and impact of the study**
- **Supplemental materials and data**

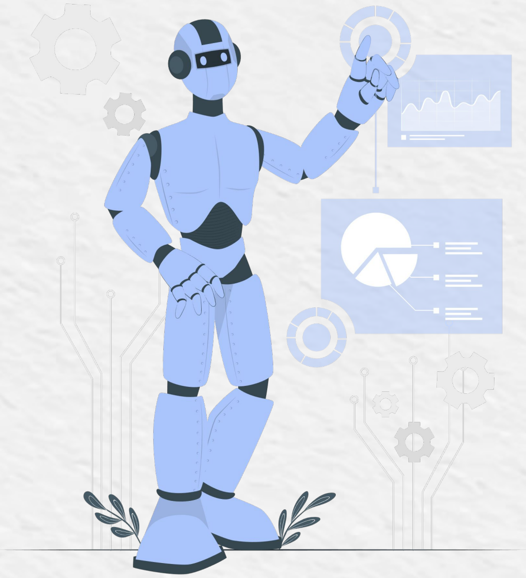
Context

Recent Advances:
GPT-4v supports multimodal input (text + images)

Huge Potential:
Automated question-answering, learning and misinformation detection.

Central Question:
Can GPT-4v interpret visual data as well as humans?

Knowledge Gap:
No equivalent evaluation for multimodal models like VLAT for humans.



Objective

Empirically evaluate GPT-4 with vision (GPT-4V) on multiple tasks:

Visualization literacy tests (VLAT)

Chart question answering

Detection of deceptive visualizations

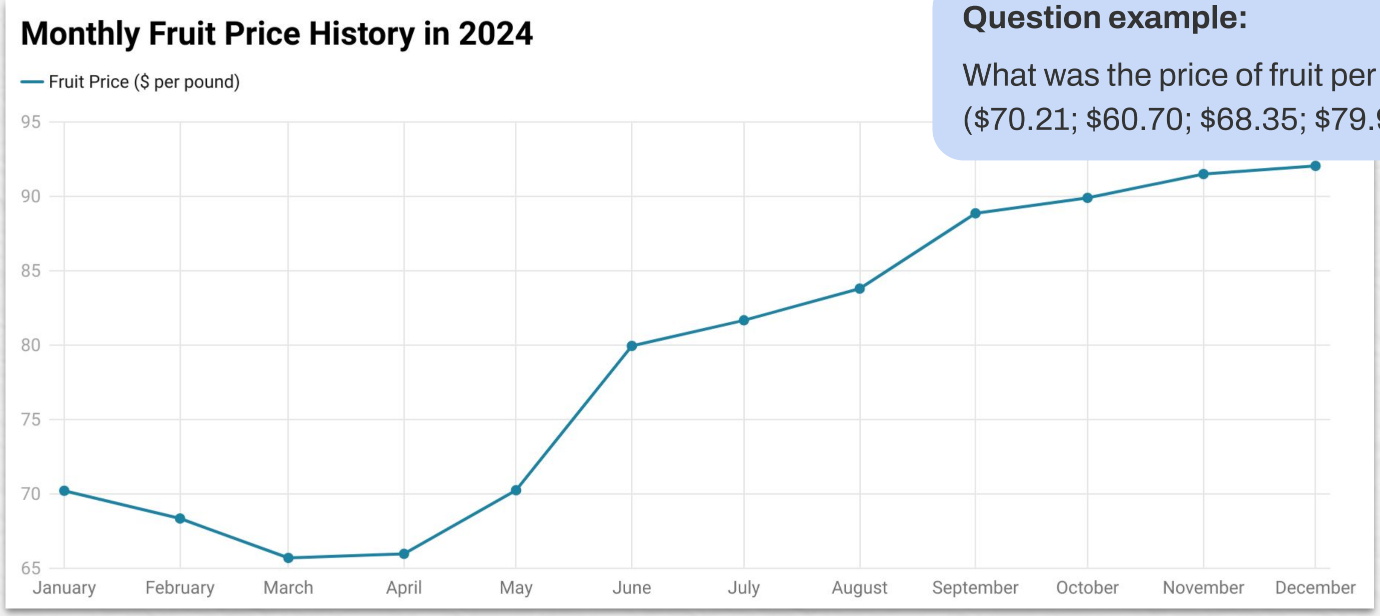
Identification of misleading or misaligned titles

Overall Evaluation Approach

Temperature 0
3 times each task
Individual task

VLAT (Visualization Literacy Assessment Test)

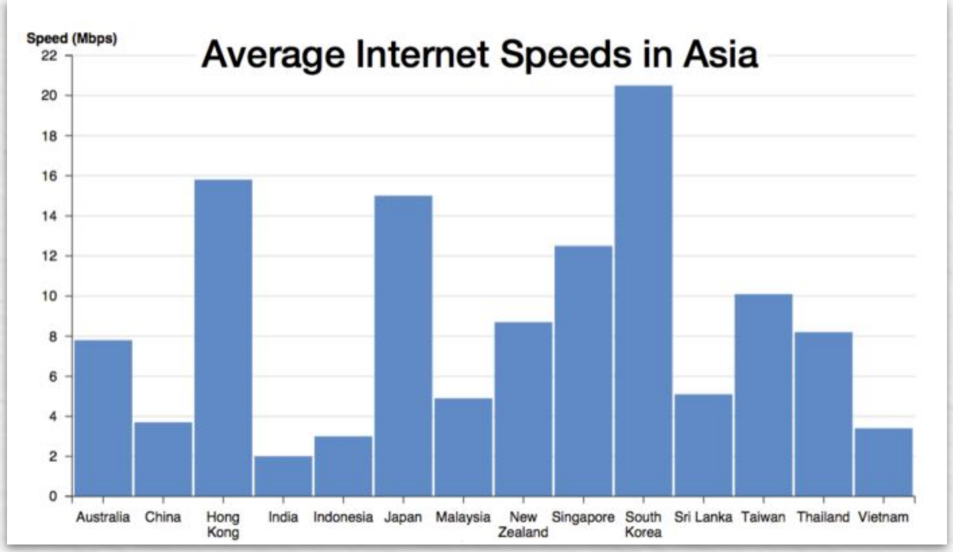
53 multiple-choice questions covering 12 chart types and 8 task types.



Question example:

What was the price of fruit per pound in January 2024?
($70.21; $60.70; $68.35; $79.95)

GPT-4's responses for 2 items on VLAT



VLAT Q6 [Easy]:
What is the average internet speed in Japan? (10 Mbps; 14 Mbps; **15 Mbps**; 16 Mbps)

Omit (1x)
14 Mbps (2x)

VLAT Q9 [Hard]:
How many countries in Asia is the average internet speed slower than Thailand? (5; 6; **7**; 8)

7 (3x)

VLAT Results

Table 2: VLAT results broken down by task type, ordered in decreasing order by percent correct.

Task type	correct	omit	incorrect	% correct
Identify hierarchy	1	0	0	100.0
Find trends	4	0	1	80.0
Make comparisons	9	1	3	69.2
Find extremum	8	2	2	66.7
Find anomalies	1	0	1	50.0
Determine range	2	0	3	40.0
Find clusters	1	0	2	33.3
Retrieve value	3	6	4	23.1

VLAT Results

GPT4v score:

19,67

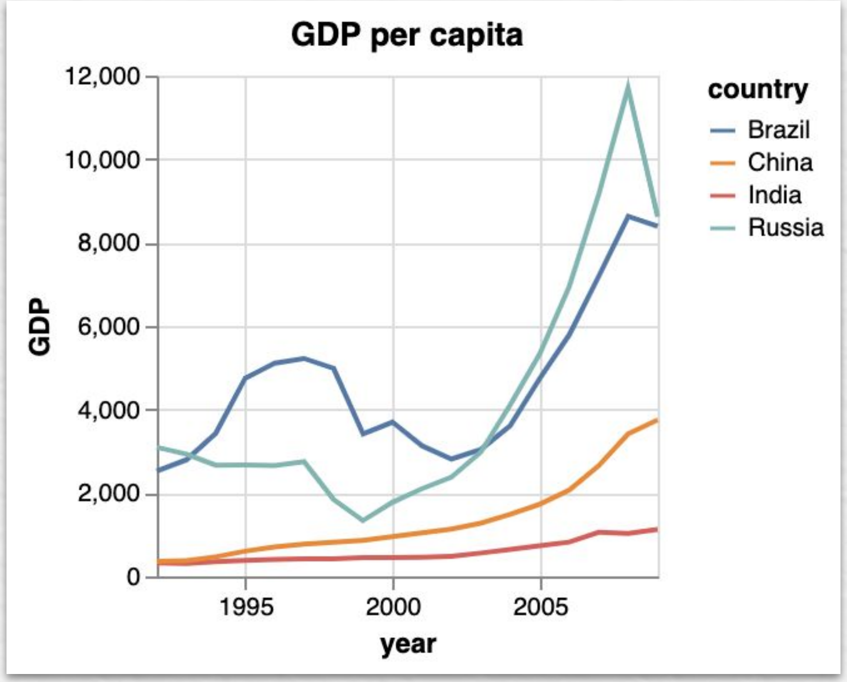
points (scoring scheme original VLAT)

Human score:	28,82
Deviation:	8,16

CQA (Chart Question Answering)

629 questions across 47 bar charts (32 simple, 8 grouped, 7 stacked) and 5 line chart

Kim et al



Question example:

Which country has the largest GDP per capita increase between year 2000 and 2005?

CQA (Chart Question Answering)

Lookup - single value retrieval
Compositional - needing several operations

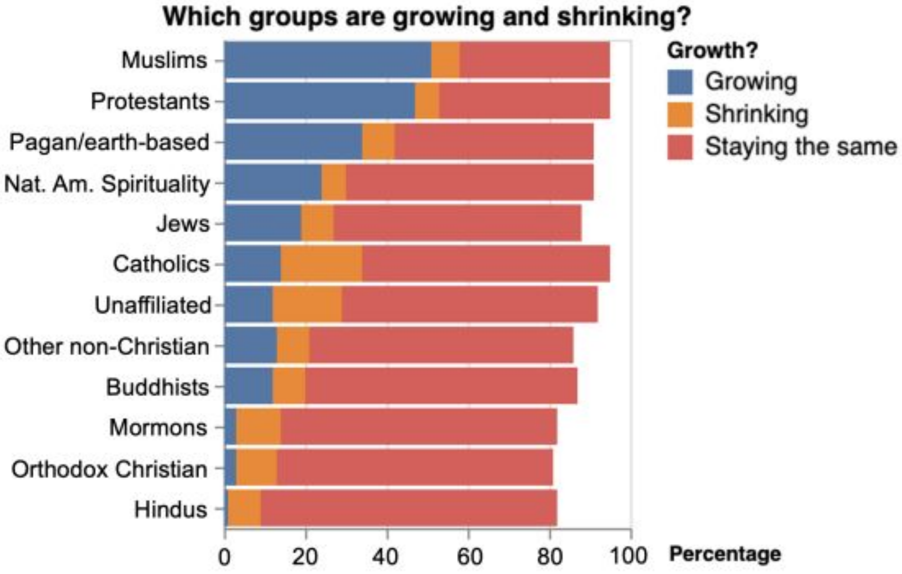
Visual - color, bar length on chart
Non-Visual - geral analysis

With Data
No Data

Table 4: Breakdown of questions in our set based on categorization of "lookup" vs. "compositional" and "visual" vs. "non-visual".

	# Questions		
	Lookup	Compositional	**Total**
Visual	52 (8%)	24 (4%)	**76 (12%)**
Non-Visual	141 (22%)	412 (66%)	**553 (88%)**
Total	**193 (30%)**	**436 (70%)**	**629**

GPT-4's responses for CQA



CQA Q85:
What is the percentage of red Protestants?

*The red portion [...] indicates the percentage that is **shrinking**. According to the provided data, 6% of Protestants are in the shrinking category.*

Fig. 3: GPT-4 consistently fails to correctly interpret the color legend on this stacked bar chart. Relevant text bolded for emphasis.

GPT-4's responses for CQA

CQA Q268:
What number of years is used as increments along the y-axis?

Kim et al.'s system:
19. I counted the number of bars.

GPT-4:
*The increments along the y-axis are in **5-year intervals**.*

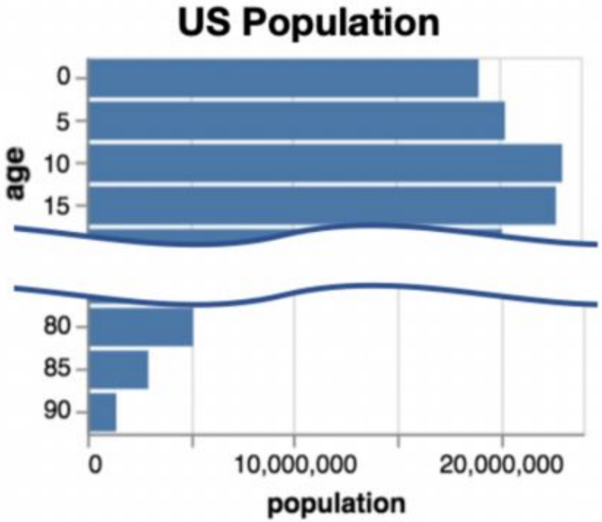


Fig. 4: GPT-4 can easily succeed on some failure cases from Kim et al., especially those related to reading axes.

CQA Results

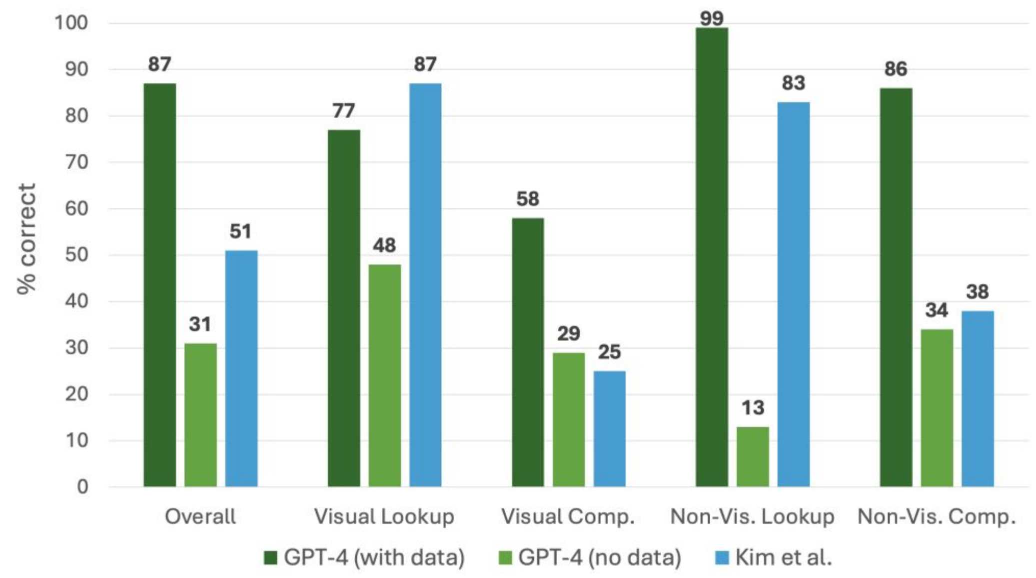


Fig. 2: CQA accuracy of GPT-4 with and without data provided, compared to the 3-stage pipeline from Kim et al.

Visual Lookup
Visual Compositional
Non-Visual Lookup
Non-Visual Compositional

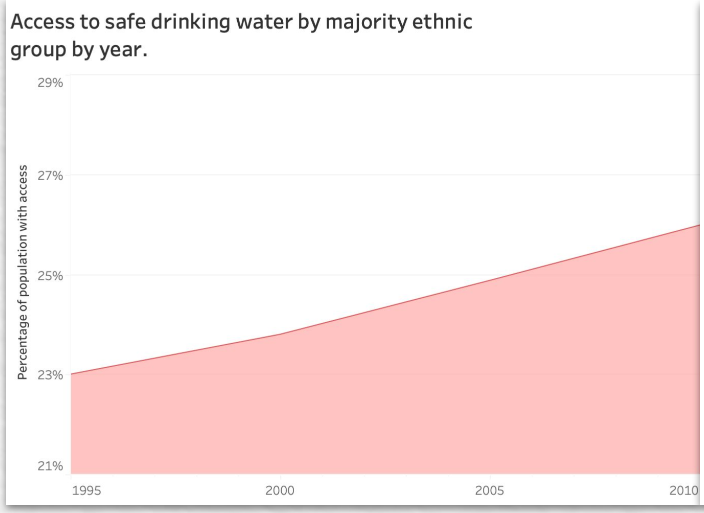
Deceptive Visualizations

6 distortion techniques, each with control and deceptive versions based on Pandey et al. (2015)

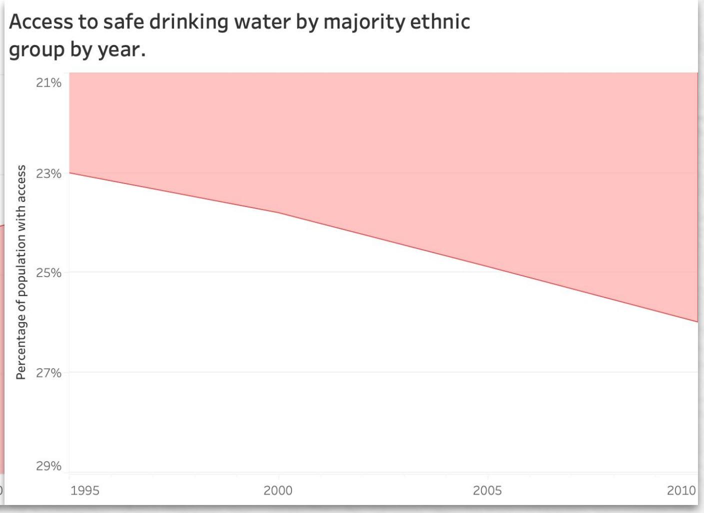
3 questions per chart:

- ❑ Interpret main message
- ❑ Detect the distortion technique
- ❑ Compare control vs deceptive

Distortion: Inverted axis



Control



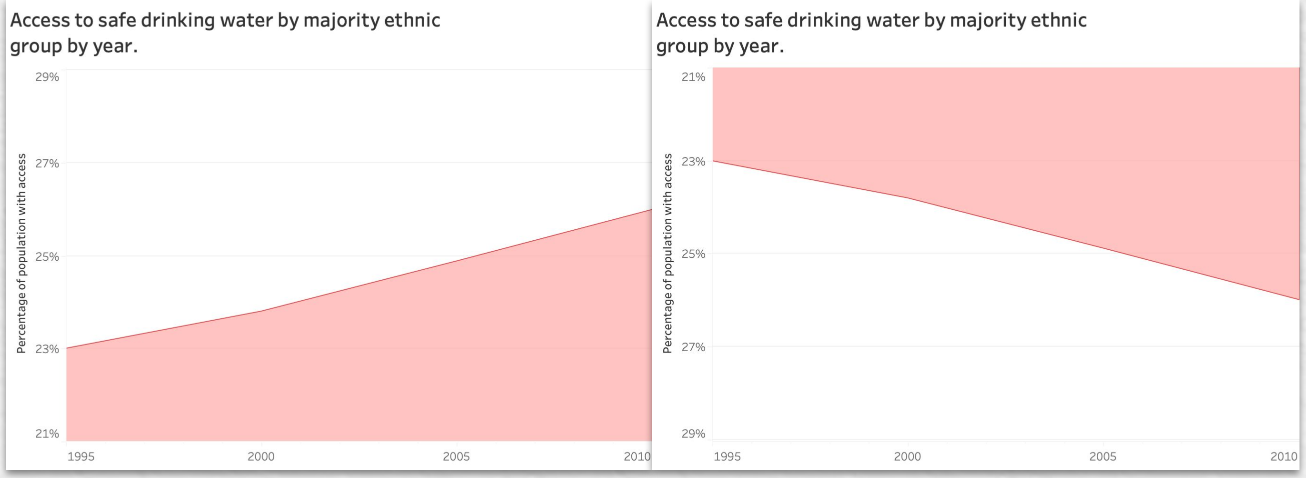
Deceptive

GPT-4’s responses for Deceptive Visualization

What can you say about the access to safe drinking water between 1995 and 2010?
Answer one of the following: Improved, Declined, Uncertain.

Improved

Declined



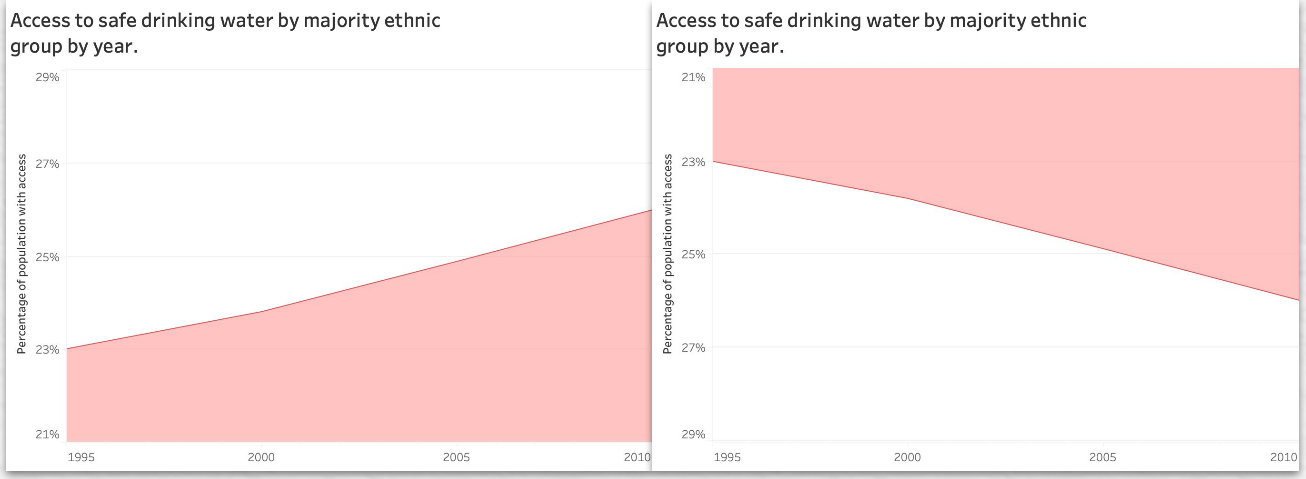
GPT-4’s responses for Deceptive Visualization

Are any misleading design tactics being used in this chart?

Truncated
y axis

Do these two charts, displayed side by side, show the same data?

No, the chart on the left shows an **increasing** trend over time, while the chart on the right shows a **decreasing** trend over the same time period.



Deceptive Visualization Results

Deceptive tactic	Fooled?	Detects (main) tactic?	Identifies chart equivalence?
Truncated axis	Yes (large effect)	Yes	Yes
Truncated axis w/ labels	Yes (small effect)	Yes	Yes
Area as quantity	No	Yes, but also detects for control	Yes*
Aspect ratio	No	No	Yes*
Inverted axis	Yes	No	No
Inverted color ramp	No	No*	Yes*

Fig. 7: Summary of results across deceptive design tactics. An asterisk (*) indicates that the response(s) contained evidence of hallucination.

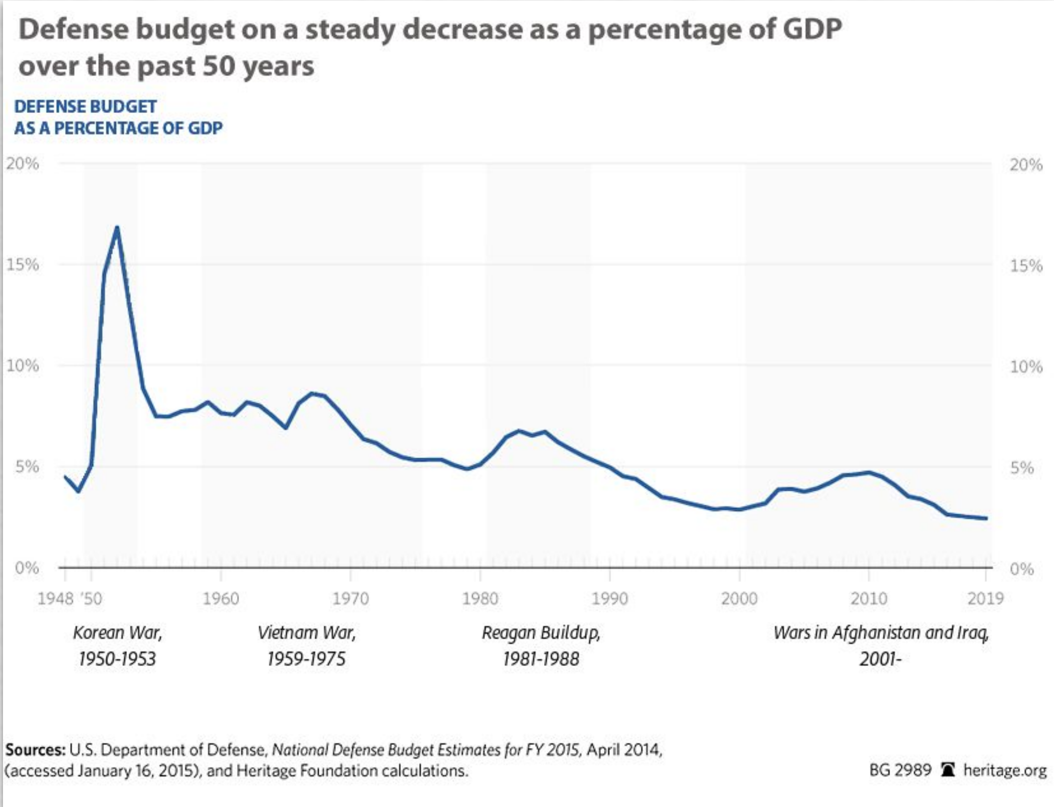
Visualizations with misaligned titles

16 visualizations total with 4 visualizations for each condition:

Control **Selective** **Miscued** **Contradictory**

2 Questions for each:
Do you find the title appropriate for this visualization?
Does the title tell the whole story?

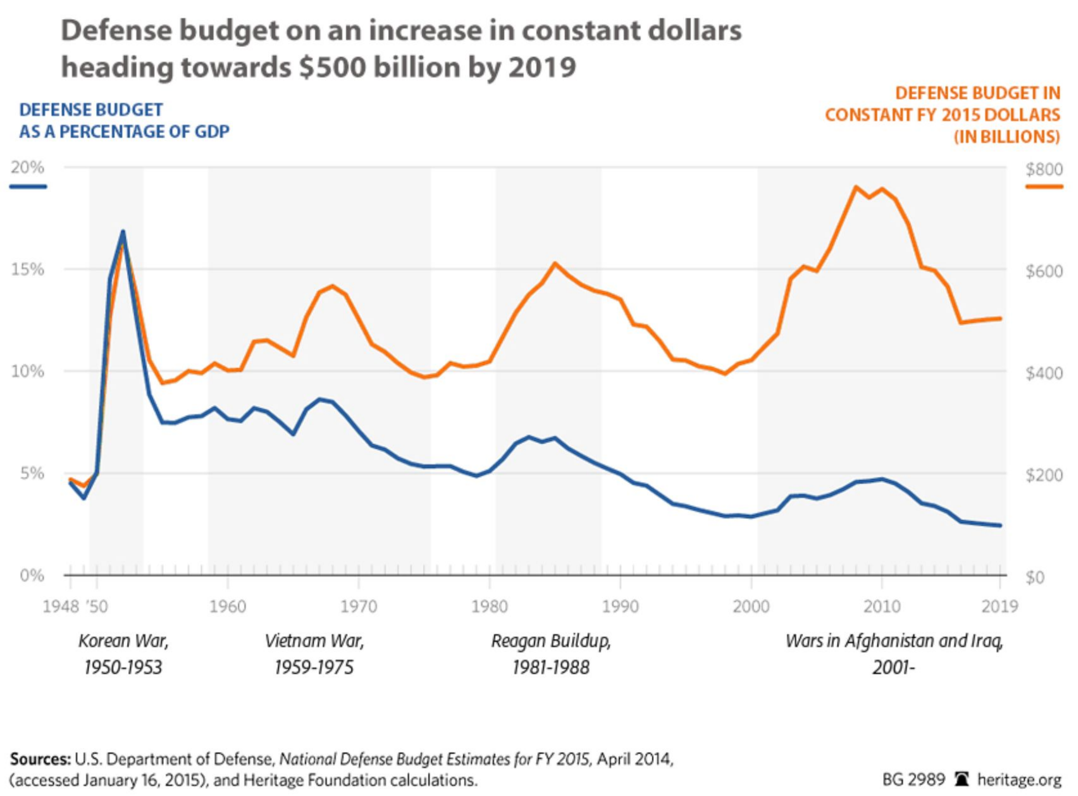
Visualizations with misaligned titles



Example of a **control** title that refers to the blue line.

Results:
Overly sensitive, often disagreed with accurate titles over minor details.

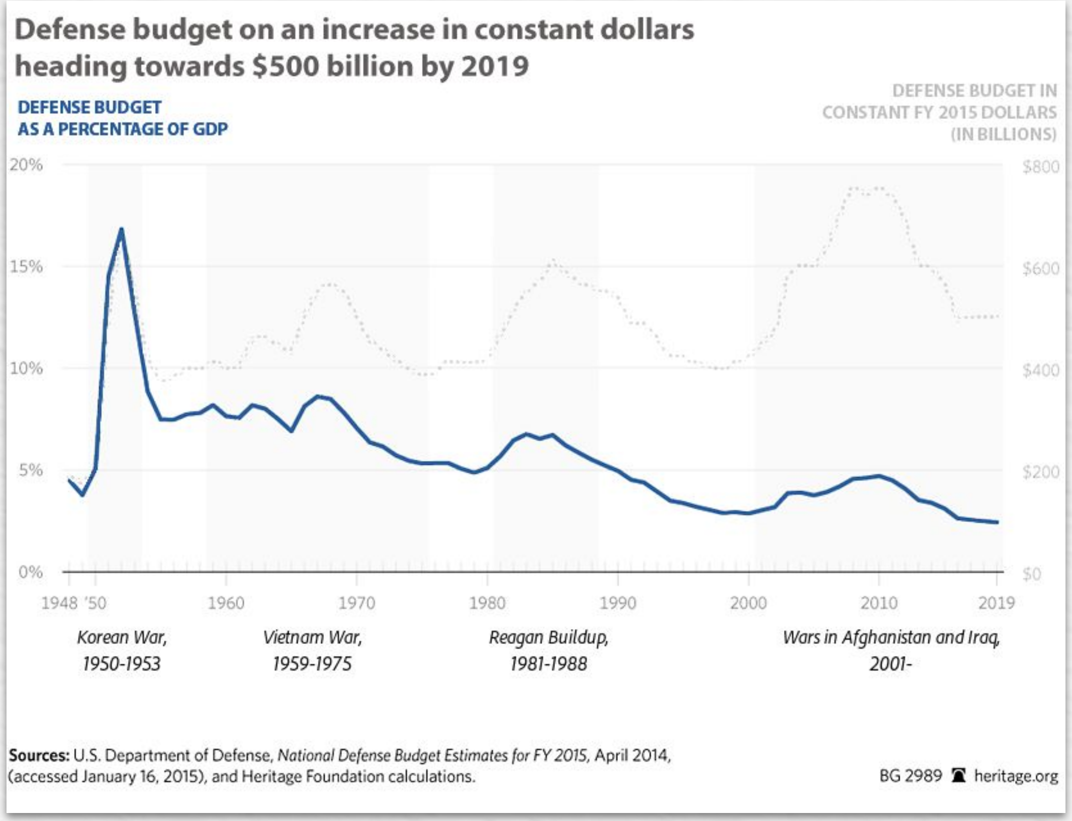
Visualizations with misaligned titles



Example of a **selective** title that refers only to the orange line.

Results:
Often accepted the title and missed the selective framing.

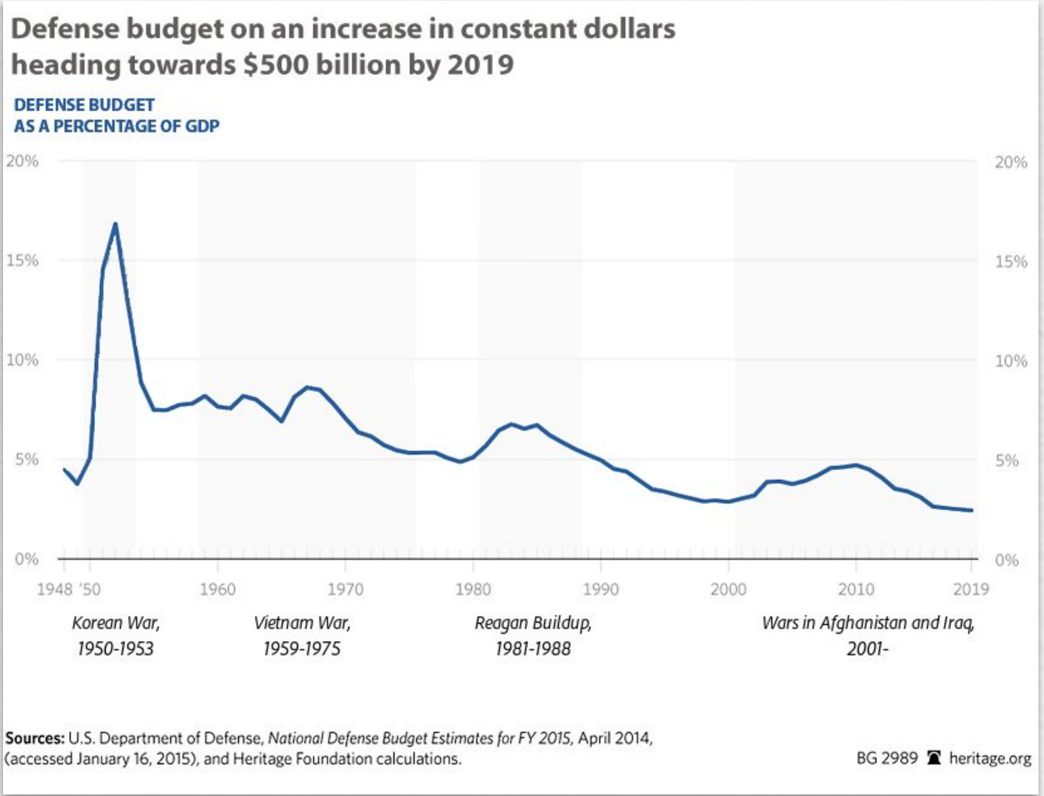
Visualizations with misaligned titles



Example of a **miscued** title that refers to the dashed line.

Results:
Detected missing information when asked about the full story but missed visual misdirection.

Visualizations with misaligned titles



Example of a **contradictory** title that refers only to the orange line, not present.

Results:
Clearly identified the contradiction between title and data

Conclusion: Strengths of GPT4v

Strong analytical and interpretive abilities

Detects global patterns and extremes

Compares values and performs multi-step calculations

Critically evaluates chart titles with well-structured prompts

Performs best when underlying data is provided

Conclusion: Weaknesses of GPT4v

Struggles to retrieve exact values from charts

Poor color distinction and spatial reasoning

Tends to focus on minor, irrelevant details

Prone to hallucination and inconsistent answers

Strongly dependent on prompt formulation

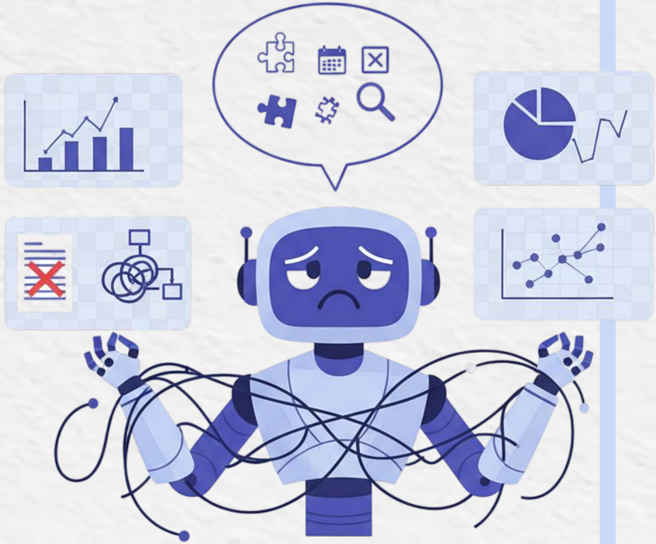
Limitations of the study

Missing visualization types and task sets

Possible benefit from prior knowledge of older datasets

Results reflect a single snapshot of a changing model

Performance depends on prompt engineering

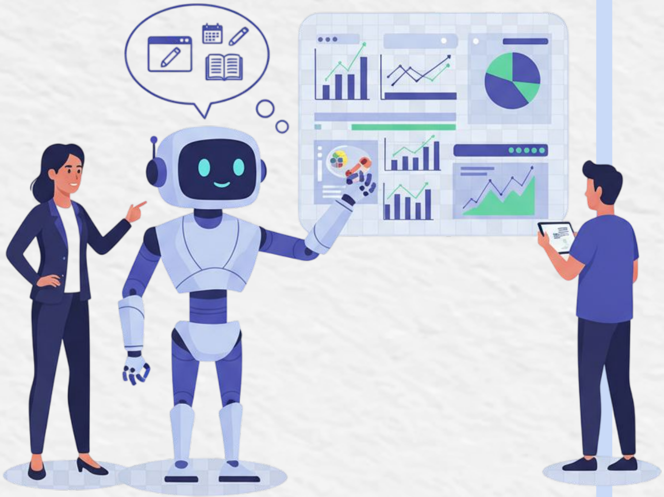


Future work

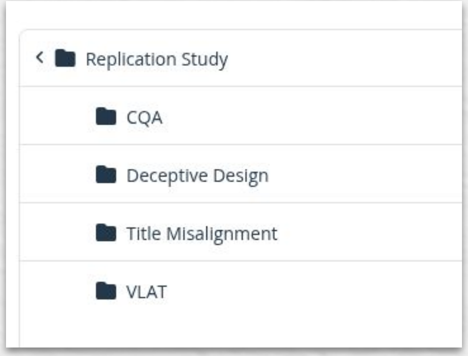
Evaluate other LLMs

Create a visualization literacy leaderboard for LLMs

Integrate into browser extensions



Our opinion



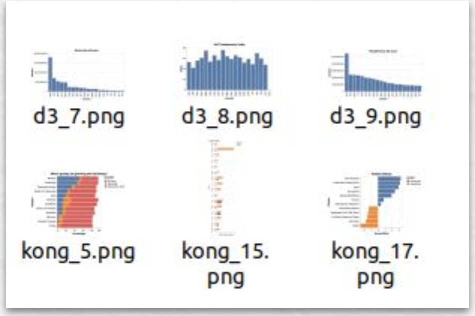
Open access to questions, results, and extra data

Information visualization to present and clarify the results

Simple reading and easy to understand

Goal of understanding GPT-4V's value in this field successfully achieved

Bad website organization



Thank you for your attention

Do you have any questions?

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